Markus Gruber, M.Sc.

Senior Machine Learning Engineer
Data Scientist

AI Infrastructure · *RAG Pipelines* · *Scientific Computing Specialist*

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Professional Profile



Analytical Senior Machine Learning Engineer & Data Scientist holding an M.Sc. in Scientific Computing with a strong mathematical foundation. Over 9 years of industry experience deploying production-grade AI pipelines and data architectures. Expertise bridges mathematical algorithm design, GPU edge acceleration (CUDA), and cloud-native RAG infrastructure (Vector DBs, PyTorch). Highly effective at productionizing complex ML models with high execution throughput.

Engineering Approach & Mindset

- **Systematic Problem Solving:** First-principles analytical approach to systematic debugging and stability optimization.
- **Quality Efficiency Focus:** Pragmatic engineering mindset balancing rapid deployment with long-term maintainability and high performance.
- **Cross-Functional Integration:** Acts as the technical glue bridging development, QA, and architecture.

Engineering Profile

TECHNICAL FOUNDATION & AI / DATA DOMAINS

AI & RAG Infrastructure	Data Engineering & Analytics	Hardware Acceleration
RAG Pipelines & Vector Databases	PySpark, DuckDB, PostgreSQL	CUDA GPU Edge Acceleration
PyTorch, TensorFlow, Keras, ONNX	Scientific Computing (M.Sc. Vienna)	Embedded Vision Pipelines
LLM Orchestration (LangChain, LangGraph)	Apache Superset & Data Pipelines	Low-Latency Inference Engines

CORE TECHNICAL SPECIALIZATION – ML & DATA SYSTEMS

- **Production AI Infrastructure:** Designing scalable RAG architectures, vector database integration, and high-throughput LLM inference backends.
- **Hardware Acceleration & Edge ML:** Optimizing deep learning models (ONNX, PyTorch) via CUDA for low-latency execution on GPU edge hardware.
- **Mathematical & Data Pipelines:** Applying numerical algorithms, distributed data processing (PySpark, DuckDB), and automated verification to data workflows.

Professional Experience

Mindbreeze GmbH – Software / Quality Engineer (C++/Python) 09/2022 – 07/2026

C++, Python, REST APIs, Linux, Docker, DevOps, Telemetry & Performance Profiling | Team: 6 Engineers

- Engineered Python/Bash automation frameworks, reducing release testing duration from 10 working days down to 3 working days.
- Optimized performance-critical backend components for enterprise architectures handling distributed data processing.
- Refactored search engine core modules and optimized index data pipelines to improve high-throughput query execution.

- Primary technical troubleshooter during release phases, resolving memory leaks and performance degradations in live systems.

Untis GmbH – Software Engineer (C++) 02/2022 – 06/2022
C++, System Refactoring, Performance Profiling, Core Desktop Modules | Team: 10 Engineers

- Refactored legacy C++/MFC core modules using Visual Studio to eliminate technical debt, optimize memory allocation, and resolve production crashes.

logi.cals – Embedded Software Engineer (C/C++) 02/2021 – 01/2022
C, C++, Embedded Systems, Safety-Critical Architectures, Deterministic Execution | Team: 6 Engineers

- Engineered deterministic C/C++ runtime execution systems (Software PLCs) and automated verification test suites for safety-standard certification across hardware platforms.

EYYES – Software Engineer (C++/Python) 04/2017 – 12/2020
C++, Python, Real-Time Execution Pipelines, CUDA, Embedded GPU Edge | Team: 8 Engineers

- Developed computer vision and deep learning pipelines (Keras, ONNX) for industrial and embedded camera systems to ensure continuous real-time edge performance.
- Engineered low-latency C++ and CUDA execution routines for real-time video stream processing on embedded GPU edge devices.
- Implemented hardware-near software components in C++ to control camera sensors, stepper motors, and SerDes interfaces.
- Optimized detection accuracy and recognition precision by training multiple TensorFlow classifiers to 99.99% accuracy through rigorous hardware-near algorithm tuning and automated benchmarking suites.

Selected Software Projects

- **Azure AI Document Analyzer & RAG Infra:** RAG processing pipeline utilizing vector embeddings, LLM orchestration, and Azure cloud integration (GitHub).
  Tech: Python, RAG, Vector DBs, LLMs, Azure
- **AWS Cloud Deep Learning Pipeline:** Deep learning vision pipeline with automated data ingestion, model evaluation, and PyTorch inference (GitHub).
  Tech: PyTorch, Computer Vision, AWS, MLOps, Python
- **Magnum C++ Game of Life:** Parallel grid simulation exploring computational cellular automata and algorithmic performance (GitHub).
  Tech: Modern C++, Algorithmic Optimization, Parallel Computing

Technical Skills

ML, AI & Data: PyTorch, TensorFlow, Keras, ONNX, RAG Infrastructure, Vector DBs, LangChain, LangGraph, PySpark, DuckDB, PostgreSQL, Apache Superset.
Core Systems & GPU: Modern C++, Python, CUDA, OpenCL, OpenMP, MPI, Linux, Docker, REST APIs.
Cloud & DevOps: Amazon AWS, Microsoft Azure, GitLab CI/CD, Git, Telemetry, Cursor, GitHub Copilot.

Education

M.Sc. Scientific Computing – University of Vienna 10/2013 – 06/2017
Thesis: Silent Data Corruption in FT-GMRES (Link, Advisor: Prof. Dr. W. Gansterer).

B.Sc. Industrial Simulation – FH St. Pölten 09/2010 – 06/2013
Thesis: Simulation of Nanocomposite Materials (Link, Advisor: Prof. Dr. T. Schrefl).

Languages & Work Status

German: Native (C2) · **English:** Full Professional Proficiency (C1/C2) · **Work Permit:** EU Citizen / Austria